

APPLICATION OF TISSUE CULTURE TECHNOLOGY FOR IMPROVEMENT OF COCONUT

One of the main objectives of the Coconut Research Institute is to increase the coconut production by supplying seedlings of improved hybrids to coconut growers. Production of new coconut varieties with conventional breeding methods is very difficult and slow process mainly due to constraints pertaining to the palm's biological characteristics and high heterozygosity (mainly seen in tall coconut varieties) as a result of cross pollination, very long breeding cycle and absent of a vegetative propagation method. Now it is very important to use modern biotechnologies like tissue culture to produce new coconut varieties and their propagation.

Production of clones through tissue culture

Clonal propagation is a technique by which a small piece of plant tissue obtained from a mother palm is cultured in an artificial media with all nutrients required for the growth of a tissue, under controlled and sterilized environment. This produces a large number of plants which are identical to the mother palm. Tissue culture technique could be used effectively to produce a large number of plantlets identical to the bearing mother palm. Genetic variations of coconut seedlings could be minimized with tissue culture techniques. If we have successful tissue culture methods, a large number of identical parental materials could be produced from a parent palm having improved characters like high yielding, resistant to pests and diseases and drought tolerance. Considering the importance in developing a perfect tissue culture technology, Coconut Research Institute of Sri Lanka and research institutions of some other counties started relevant research programmes about two decades ago. Initially tender leaf, and immature inflorescences were used in clonal propagation research. However results of experiments received by using the above materials were unsatisfactory and then immature embryos and plumules were used for research. Generally each seed nut produces only one seedling in germination as it has only one embryo. However, research has confirmed that using the tissue culture technology, a large number of plantlets could be produces from one embryo. The development of clonal technology to produce a large number of clonal plantlets from a single embryo could be considered as a major achievement in clonal propagation in coconut. However, embryo is produced as a result of cross pollination; plantlets produced from these embryos cannot be referred as real clones. CRI continued research activities to

identify a suitable tissue for clonal propagation technology. The suitability of different tissues obtained from immature inflorescence was re-assessed. A thorough investigation was carried out, as the initial step, of this assessment for the initiation and development of coconut inflorescence by using tissue culture technologies. Based on these findings, recent experiments reveals that the ovary obtained from immature female flower is the most suitable tissue for clonal propagation.

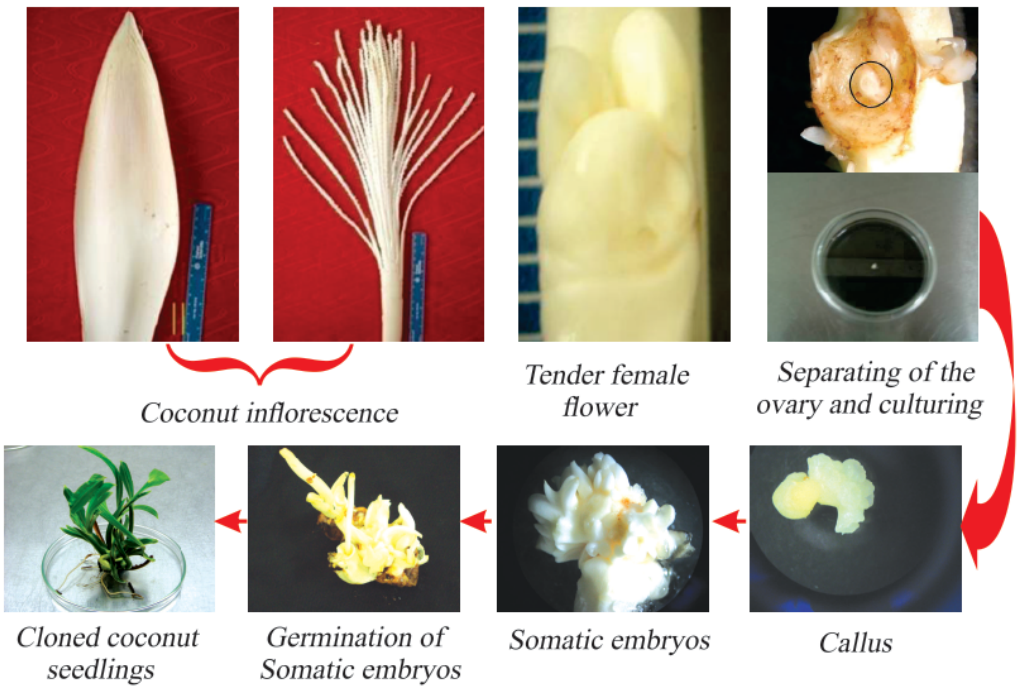


figure1: Production of clones using immature ovary culture

It is revealed that immature embryos (regenerating materials) or other tissues obtained from mother palm produce clonal plants through somatic embryo genesis. When the regenerating materials are grown in nutrient mixture initially they produce a mass of undivided cells referred as callus. It has been possible to increase the number of cells in the callus with cell multiplication process. Then some cells of callus multiplies and develop into somatic embryos (Somatic seeds). These somatic seeds germinate and produce clonal plantlets.

It is important to identify optimum conditions (composition of nutrient medium and physical factors) required for different stages of regeneration process including callus initiation, callus multification, initiation of somatic embryos and clonal plants. The best results could be obtained by deciding optimum concentration of different ingredients (mineral salts, vitamins, amino acids, hormones, energy sources and activated carbon) of the growing media

Further experiments are carried out to develop protocols to obtain clonal plants from the germination of mature embryos. One of the steps of this process is to identify the most responding embryos out of the embryos collected from different mother palms.

In the clonal plant production process different stages are being studied further in order to maximize clonal plant production.

Production of pure line coconut

Very long time involves in the conventional breeding process is considered as a main barrio in the production of new coconut hybrids. For the production of hybrids if pure coconut lines could be used the time requirement could be reduced. However, it very difficult to obtain pure coconut lines with conventional methods. Experiments were carried out to test the possibility of using tissue culture technology for the production of homozygous or pure lined palms.

Coconut pollen grains mean haploid cells, found in large numbers in anthers of male coconut flowers. Anthers with a large number of pollen grains are germinated in nutrient media to produce haploid plants (produced from pollen grains). These plants could be used for the production of pure coconut lines by pairing their chromosomes. The homozygous palms thus obtained could be hybridized to produce new coconut hybrids.

With these objectives experiments were carried out few years ago, on the germination of coconut anthers which brought about very successful results. The above experiments helped to identify most suitable anthers with germination capacity and the most suitable growing media (composition and physical factors) to produce large number of homozygous plants and to find out most suitable methods to induce anther embryo genesis. This also helps to identify the correct method of placing anthers in the media and maintain their density. The above experiments carried out, produced a group of homozygous palms from the germination of anthers in test tubes, for the first time in the tissue culture history.

These new findings will be highly useful in developing coconut breeding activities in future.

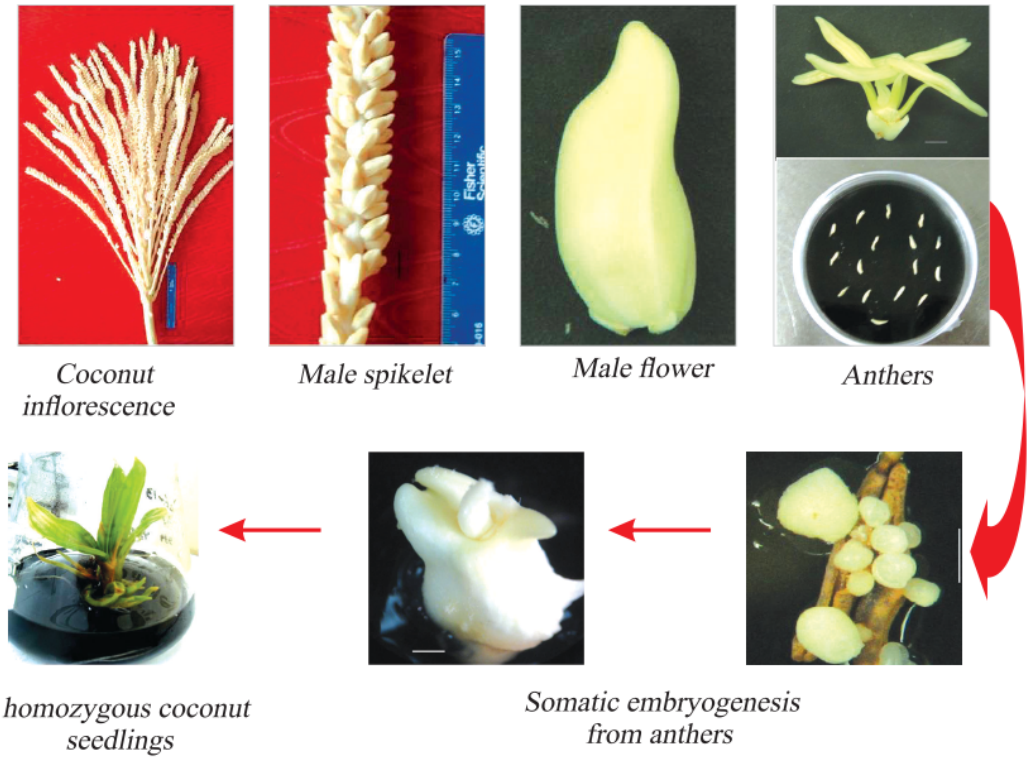


figure 2: Production of homozygous coconut seedlings using anther culture

Mature embryo culture

In mature embryo culture, an embryo separated from a mature nut is germinated and developed into a plant by growing it in a sterilized artificial nutrient media. In this method plant nutrients normally supplied by the endosperm of the nut are provided through the artificial nutrient media to develop embryo into a plant. To develop a full grown plant, this process normally takes 6-8 months and produces a single plant from each embryo. This method is mostly important to exchange germplam materials between counties. The major limitation in the production of new coconut verities is the availability of very limited number of coconut varieties in Sri Lanka.

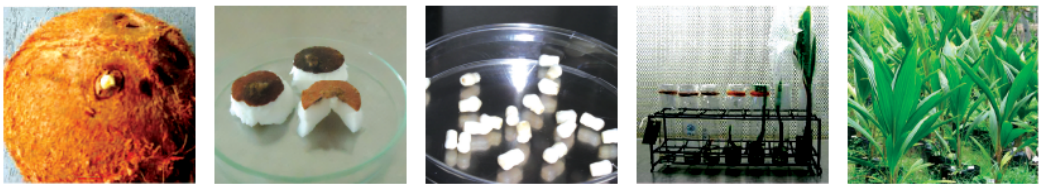


figure 3: Different stages of embryo culture

The best answer to this problem is bringing new coconut varieties from other coconut growing counties. Importation of mature nuts from other countries for germplam exchange is a risky operation due to the facts that the possibility of spreading various diseases and coconut pests in this country. Under this circumstance, importation of coconut from other countries is prohibited. Therefore, the safest method for germplam exchange between coconut growing countries is the embryo culture method. As coconut embryos are very small (0.5 cm in length) they can be transported in a small bottle containing disinfected media. In this way, a large number of embryos could be brought in a small area. In addition, germination of embryos are carried out under sterilized conditions and the possibility of disease and pest incidents are minimal.

During the period between yeas of 2002 -2004, 23 coconut varieties were imported using the above technique from three countries, India, Papua New Guinea and Ivory Coast and 18 plants were established in the field. In addition in 2010 nine embryos of new coconut varieties were brought to the country from Ivory Coast and plants obtained from them were transplanted in the field. Arrangements are made to utilize foreign coconut varieties for the production of new coconut verities.

In addition, experiments are under way to conserve mature embryos with the objectives of helping to conserve germplams. The conditions required for short term and medium term conservation have been observed now. Germinating media used facilitates maintaining the minimum growth of embryo and enhance the growth of embryo when conditions are desirable. Basic methods have been developed to place coconut embryos in liquid nitrogen to conserve coconut germplams for a longer period. Further experiments are being carried out on this line.

The embryo culture technology is presently used to propagate special coconut types known as Dikiri coconut. It has been identified that in a Dikiri coconut palms nuts bearing are with 2% - 20% endosperm with double recessive genes. Dikiri nuts cannot be germinated in the field and produce seedlings. Dikiri embryos could be germinated in a tube and produce palms with embryo culture technology with capacity of producing 100% Dikiri nuts. Coconut palms bearing Dikiri nuts are found in the Southern Province especially thriving in the Weligama area. People in these areas used Dikiri coconut to produce different sweets. The cost of each Dikiri coconut is within the range of Rs.150/= to Rs.200/=. In order to increase the income from home gardens, Coconut Research Institute introduced a programme to produce and distribute Dikiri plants to coconut small holders. Dikiri demonstration

plantations were established in the experiment center in Middniya, Dikiri plants were issued to growers at concessionary price.

Acclimatization of plants grown in tubes or hardening

Acclimatization is a process of training small plants produced in tubes as a result of immature embryo culture, anther culture and mature embryo culture to expose to the external environment and bring them to field planting conditions which is referred as hardening. The small planted cultivated in tubes under specific conditions subject to dehydration due to poor regulation of stomata at the initial stage. Hardening process should be carried out carefully as plants are susceptible to pathogens at this stage. At the initial state plants are exposed to low sunlight and the environment with high humidity. Then plants are exposed to external environment gradually. This process requires few months.



figure 4: Different stages of acclimatization

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